TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Dimensional Stability

Shrinkage and growth measured between bench marks through standard washing — relaxation, felting and progressive shrinkage, and the tolerance that decides pass or fail.

TE-3202 · QUALITY EVALUATION OF FABRIC AND APPAREL SESSIONAL · EXPERIMENT 12 · LEVEL 3 TERM 2

1. Aim and theory

Aim: to determine the dimensional change of a fabric after washing, following the principles of ISO 3759 (preparation, marking and measuring of specimens) and ISO 6330 (domestic washing and drying procedures), to calculate warp and weft dimensional change percentages, to distinguish the mechanisms of shrinkage, and to judge the result against a buyer tolerance.

1.1 Why fabrics change size

A fabric leaves the loom and the finishing range full of stored strain. Yarns were held under tension in weaving, stretched in stentering to a commercial width, and dried in that extended state; the structure the customer buys is therefore wider, longer or straighter than the structure the fibres would choose if released. Washing releases it. Water swells the fibres, heat and agitation give the molecules and yarns mobility, tensions relax, yarns crimp more and the fabric contracts toward its relaxed dimensions. This first, large, essentially one-way movement is **relaxation shrinkage**. It is not a defect of the fibre but of the finishing: a fabric properly relaxation-shrunk in finishing (for example by a compressive shrinkage process on cotton) has already made this movement before sale and changes little in the customer's wash.

Wool adds a second mechanism. Its scaly surface makes fibres migrate preferentially root-first under wet agitation; the fabric entangles irreversibly, thickens and shrinks wash after wash. This is **felting shrinkage**, and unlike relaxation it is progressive — it does not finish after the first wash. A third pattern, **progressive shrinkage** in other fibres, describes any slower continued contraction over repeated washes as residual strains and yarn rearrangements work themselves out, so a complete study measures after several cycles (three washes in this exercise), not one. The opposite sign also occurs: some fabrics **grow** — knits stretched in wear and finishing can extend, and some constructions relax into length. The sign convention handles both:

$$\text{Dimensional change (\%)} = (\text{final length} - \text{original length}) / \text{original length} \times 100$$

A negative result is shrinkage; a positive result is growth (extension). Results are reported for each direction separately to one decimal place or better, with the full washing and drying procedure named — a shrinkage figure without its wash procedure is an opinion, because severity lives in the procedure: temperature, detergent, mechanical action, load and drying method all move the result.

1.2 The bench-mark method: ISO 3759 and ISO 6330

ISO 3759 standardises how the change is measured. On a conditioned specimen, pairs of **bench marks** are placed at a known spacing — 350 mm in this sheet — in both directions, three pairs in each direction, positioned away from edges and selvedges so that local distortion does not dominate. Indelible marking that survives washing, measured with a rigid rule or calibrated scale on a flat, relaxed specimen, keeps the measurement error a small fraction of the change being claimed. ISO 6330 then standardises the washing itself — machine type, programme, temperature, detergent dose, total load mass with ballast, and the drying method (line dry, flat dry, tumble dry and their variants) — so that "washed three times" means the same insult in every laboratory. Specimens are conditioned in the standard atmosphere (65% RH, 20 °C) before marking and again before final measurement, laid flat without tension, and allowed to relax fully; measuring a warm, freshly tumbled specimen pulled straight by hand manufactures whichever answer the operator expects.

1.3 Tolerances and garment consequences

Buyers do not demand zero change — they specify a tolerance band per direction, commonly of the order of ±3% in the warp and ±2–3% in the weft for wovens (knits are usually allowed more), because garment patterns are cut with the expected behaviour in mind and small, known changes are absorbed in seam allowances and ease. Failure is directional and practical: warp shrinkage shortens trousers and sleeves and puckers seams whose sewing thread does not shrink with the cloth; weft shrinkage tightens the body and can close a garment out of its size; unequal change between a shell fabric and its lining or interlining twists and blisters the garment. A result marginally outside tolerance is therefore a genuine commercial failure, not pedantry — the garment that leaves the customer's first wash is the garment the customer judges.

Dry cleaning deserves the note this sheet gives it: dimensional change is also measured after dry cleaning where the care label permits it, using the appropriate solvent procedure, since some finishes and interlinings move in solvent while stable in water (and vice versa). The report always pairs the figure with the medium: "−3.1% warp after 3 × ISO 6330 washes, line dried" is a result; "shrinks 3%" is not.

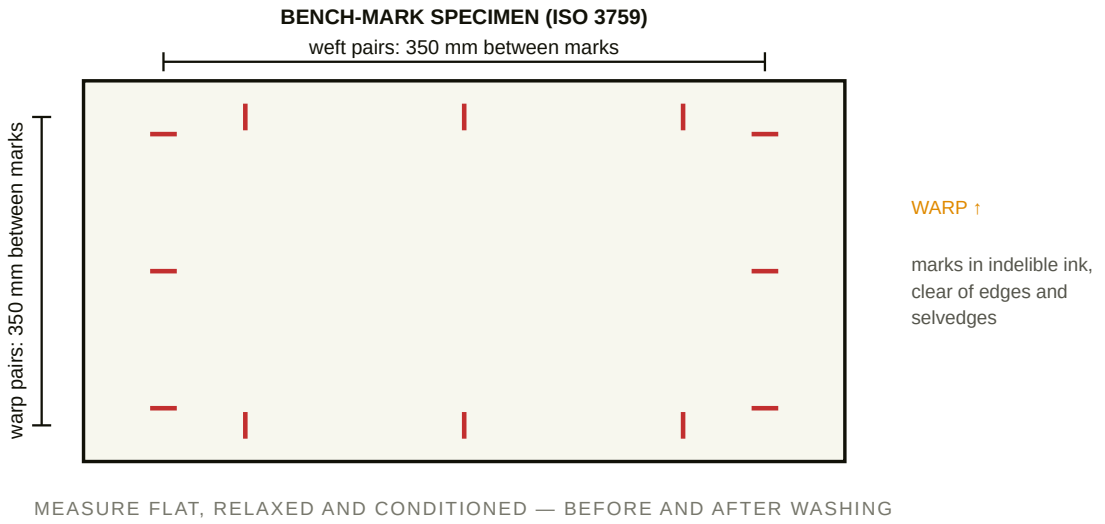


Figure 1 — A marked specimen: three pairs of bench marks across the warp direction and three down the weft direction, each pair 350 mm apart. Six distances before washing and the same six after washing give the directional changes with local distortion averaged out.

2. Apparatus and procedure

- **Apparatus:** washing machine and procedure to ISO 6330 with specified detergent, ballast and drying line; indelible marker and marking template; rigid steel rule or calibrated measuring scale (mm); flat measuring table; conditioning facilities (65% RH, 20 °C); balance for machine load if required by the procedure.
- **Preparation (ISO 3759):** cut specimens clear of selvedges and creases, condition them in the standard atmosphere (65% RH, 20 °C), lay flat without tension, and mark three pairs of bench marks in the warp direction and three in the weft direction, each pair exactly 350 mm apart, the pairs spread across the specimen as in Figure 1. Measure and record all six distances.
- **Washing (ISO 6330):** wash the specimens with the specified ballast load, programme, temperature and detergent for one cycle; dry by the specified method. Repeat to a total of three washes for this exercise, drying the same way each time.
- **Final measurement:** condition the dried specimens again in the standard atmosphere, lay them flat and relaxed — smoothed by hand only, never stretched — and re-measure all six distances at the same marks.
- **Calculation and reporting:** average the three warp distances before and after, and the three weft distances before and after; calculate percentage change for each direction with its sign. Report the procedure in full (standard, programme, temperature, wash count, drying method), the two percentages, and the verdict against the tolerance stated in the exercise or by the buyer.

3. Worked calculation

Mean bench-mark distances (three pairs averaged in each direction), original spacing 350 mm, after three ISO 6330 washes with line drying:

DIRECTION	ORIGINAL (MM)	AFTER 3 WASHES (MM)	CHANGE (MM)	DIMENSIONAL CHANGE (%)
Warp	350	339	339 − 350 = −11	−11 / 350 × 100 = **−3.14%**
Weft	350	344	344 − 350 = −6	−6 / 350 × 100 = **−1.71%**

Warp: (339 − 350) / 350 × 100 = −3.14% · Weft: (344 − 350) / 350 × 100 = −1.71%

Both directions shrink, warp more than weft — the common woven pattern, since the warp carried the greater weaving and finishing tensions and therefore holds the greater stored relaxation. Judged against a typical buyer tolerance of ±3% warp and ±2% weft: the weft result (−1.71%) passes comfortably, but the warp result (−3.14%) lies just outside its −3% limit — a **marginal warp failure**. Marginal is still failure: across a bulk production the distribution of results straddles the limit, garments at the tail shorten visibly, and the correct laboratory action is to report the fail plainly, with the procedure, and to recommend the remedy at finishing — more complete relaxation or compressive shrinkage (sanforising) targeted at the warp — rather than to re-wash until a kinder number appears.

> **RESULT**
>
> After 3 washes: warp −3.14%, weft −1.71%. Against ±3% warp / ±2% weft tolerances: weft passes, warp marginally fails. Predominantly relaxation shrinkage, concentrated in the direction of greatest finishing tension.

4. Precautions

- Use marks that genuinely survive three washes and measure at the centres of the marks each time. A faded, spread mark adds millimetres of doubt to a test whose verdict may turn on half a percent — 1.75 mm on a 350 mm gauge.
- Condition before both measurements and measure flat and relaxed. A specimen measured warm from the dryer, or pulled smooth against its will, reports the operator's hands, not the fabric.
- Follow the named ISO 6330 procedure exactly — load, ballast, detergent, temperature, drying. Substituting tumble drying for line drying is a different, harsher test and must never be reported under the original procedure's name.

- Complete all specified washes before judging. Progressive and felting shrinkage are defined by their behaviour over cycles; a one-wash pass does not clear a fabric whose second and third washes keep moving.
- Average the three pairs in each direction and inspect the individual pairs. One pair moving very differently from its neighbours signals a local distortion or a marking error to investigate, not to hide inside a mean.
- Record the sign correctly and report growth as growth. A plus result quietly rewritten as a small shrinkage reverses the garment fault — pieces grow out of size instead of shrinking out of it.

Practice (viva preparation)

1. Define relaxation shrinkage and felting shrinkage, and say which is progressive.
2. Warp marks measure 350 mm before and 339 mm after washing. Find the dimensional change.
3. Weft marks measure 350 mm before and 344 mm after washing. Find the dimensional change.
4. Judge the two results of Questions 2 and 3 against tolerances of ±3% warp and ±2% weft.
5. Why does a shrinkage report always name its washing procedure and number of washes?
6. What do ISO 3759 and ISO 6330 each standardise?
7. A fabric grows by +1.5% in length. What happens to a garment cut exactly to size, and how is the sign shown in the report?

Answer key

1. Relaxation shrinkage is the release, on first wetting, of strains stored in the fabric during weaving and finishing — yarns crimp up and the fabric contracts toward its relaxed state; it is large at first and essentially complete once the stored strain is released. Felting shrinkage is the irreversible entanglement of wool fibres caused by their scale structure under wet agitation. Felting is the progressive one: it continues, wash after wash, for the life of an untreated wool garment.
2. (339 − 350) / 350 × 100 = −11 / 350 × 100 = −3.14%. The negative sign denotes shrinkage of 3.14%.
3. (344 − 350) / 350 × 100 = −6 / 350 × 100 = −1.71% — shrinkage of 1.71%.
4. Weft −1.71% lies inside its ±2% band — pass. Warp −3.14% lies outside its ±3% band by 0.14 of a percentage point — fail, marginally. The fabric as a whole therefore fails the specification, with the recommendation directed at warp relaxation in finishing.
5. Because the result is a product of the procedure: temperature, detergent, action, load, drying method and cycle count all change the movement, and mechanisms such as felting declare themselves only over repeated washes. The same fabric can honestly show −1% after one gentle wash and −5% after

five standard washes. Without the procedure named, two laboratories cannot be compared and the figure cannot be enforced.

6. ISO 3759 standardises the specimen side — preparation, conditioning, placement of bench marks and how dimensional change is measured. ISO 6330 standardises the treatment side — the domestic washing and drying procedures (machine, programme, detergent, load and drying methods) to which the marked specimens are subjected.
7. The garment becomes longer than its pattern — sleeves and bodies drop by 1.5% of their length — the mirror-image fault of shrinkage, causing fit and matching problems of its own. It is reported with a positive sign, +1.5%, under the same formula: (final − original) / original × 100. The sign is part of the result and is never suppressed.
